MIT 6.1200J
Problem Set 1

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Problem 2

(a) Use a truth table to verify that the propositional formula $(P \wedge \neg Q) \vee (P \wedge Q) \equiv P$ is equivalent to P.

Answer.

P	Q	$\neg Q$	$P \wedge \neg Q$	$P \wedge Q$	$(P \wedge \neg Q) \vee (P \wedge Q)$
T	T	F	F	T	T
T	F	T	T	F	T
F	T	F	F	F	F
F	F	T	F	F	F

$$(P \wedge \neg Q) \vee (P \wedge Q) \equiv P$$

(b) Prove that $A = (A-B) \cup (A \cap B)$ for all sets A, B, by showing $x \in A \iff x \in (A-B) \cup (A \cap B)$ for all elements x. How does this relate to part (a)?

Answer.

Let x be arbitrary. Then,

$$\begin{aligned} & x \in (A - B) \cup (A \cap B) \\ \iff & x \in (A - B) \ \vee \ x \in (A \cap B) \\ \iff & (x \in A \wedge x \notin B) \ \vee \ (x \in A \wedge x \in B) \\ \iff & x \in A \wedge (\neg(x \in B) \vee x \in B) \\ \iff & x \in A \wedge \text{True} \\ \iff & x \in A \end{aligned}$$

Hence,

$$x \in (A - B) \cup (A \cap B) \iff x \in A$$

Therefore,

$$A = (A - B) \cup (A \cap B)$$

Relation to part (a): This identity is an instance of the propositional equivalence

$$(P \wedge \neg Q) \ \vee \ (P \wedge Q) \equiv P$$

where P corresponds to $x \in A$ and Q corresponds to $x \in B$.

Problem 3

(a) The predicate sixish(n) is defined as "n is a natural number whose first digit is 6."

Answer.

$$\text{sixish}(n) := \exists k. \Big(6 \cdot 10^k \leq n \land n < 7 \cdot 10^k\Big)$$

(b) "There are exactly two natural numbers n such that $Q(n)$ is true."

Answer.

$$\exists x\ \exists y. \Big(x \neq y \land Q(x) \land Q(y) \land \forall z. \big(Q(z) \implies (z = x \lor z = y)\big)\Big)$$

(c) Goldbach's Conjecture: "Every even integer greater than 2 can be written as the sum of two primes."

Answer.

$$\forall n. \Big(\big(n > 2 \land \exists k.\, n = 2k\big) \implies \exists p\ \exists q. \big(\text{isPrime}(p) \land \text{isPrime}(q) \land p + q = n\big)\Big)$$

(d) "Every prime larger than 100 is sixish."

Answer.

$$\forall n. \Big(\big(\text{isPrime}(n) \land n > 100\big) \implies \text{sixish}(n)\Big)$$

(e) Is the predicate $R(n) := \exists b.\, n = 2b + 5$ equivalent to isOdd(n)?

Answer.

No, they are not equivalent. We show $R(n) \not\iff \text{isOdd}(n)$ by counterexample.

Counterexample: Take $n = 1$.

- isOdd(1) is **true**: set $m = 0$, then $2(0) + 1 = 1$. ✓
- $R(1)$ is **false**: $R(1)$ requires $\exists b \in \mathbb{N}$ such that $2b + 5 = 1$, i.e., $2b = -4$. Since $b \in \mathbb{N}$, no such b exists.

Therefore the predicates are **not equivalent**.

Problem 4

(a) We prove the following statement by induction on n.

Inductive Hypothesis $P(n)$: For any n numbers $x_1, x_2, \ldots, x_n \in \mathbb{R}$, if $Q(x_1 \cdot x_2 \cdots x_n)$ is true, then at least one of $Q(x_1), Q(x_2), \ldots, Q(x_n)$ is true.

Base Case ($n = 1$): Suppose $Q(x_1)$ is true. Then trivially, $Q(x_1)$ is true. So $P(1)$ holds.

Inductive Step: Assume $P(n)$ holds for some $n \geq 1$ (inductive hypothesis). We want to show $P(n+1)$: that for any $x_1, \ldots, x_{n+1} \in \mathbb{R}$, if $Q(x_1 \cdot x_2 \cdots x_{n+1})$ is true, then at least one of $Q(x_1), \ldots, Q(x_{n+1})$ is true.

Assumption: Suppose $Q(x_1 \cdot x_2 \cdots x_{n+1})$ is true.

Goal: Show at least one of $Q(x_1), \ldots, Q(x_{n+1})$ is true.

Let $y = x_1 \cdot x_2 \cdots x_n$, so that

$$x_1 \cdot x_2 \cdots x_{n+1} = y \cdot x_{n+1}.$$

Since $Q(y \cdot x_{n+1})$ is true, the given property of Q tells us that at least one of $Q(y)$ or $Q(x_{n+1})$ is true. We consider two cases:

- **Case 1**: $Q(x_{n+1})$ is true. Then we are done.
- **Case 2**: $Q(y) = Q(x_1 \cdot x_2 \cdots x_n)$ is true. By the inductive hypothesis $P(n)$, at least one of $Q(x_1), Q(x_2), \ldots, Q(x_n)$ is true.

In either case, at least one of $Q(x_1), \ldots, Q(x_{n+1})$ is true. This completes the inductive step.

By induction, $P(n)$ holds for all integers $n \geq 1$. ■

(b) We choose $Q(x) :=$ "x is irrational."

With this choice, part (a) states: if $x_1 \cdot x_2 \cdots x_n$ is irrational, then at least one of $x_1, \ldots, x_n$ is irrational, which is exactly Theorem 1.

However, to apply part (a), we must first verify that Q satisfies the required property from part (a). This gives us the following lemma:

Lemma: For any $x, y \in \mathbb{R}$, if $x \cdot y$ is irrational, then at least one of x or y is irrational.

(c) **Proof of Lemma** (by contradiction):

Suppose for contradiction that $x \cdot y$ is irrational, but neither x nor y is irrational. Then both x and y are rational, so we may write

$$x = \frac{a}{b}, \qquad y = \frac{c}{d},$$

for some integers a, b, c, d with $b \neq 0$ and $d \neq 0$. Then

$$x \cdot y = \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}.$$

Since a, b, c, d are integers, ac and bd are integers, and $bd \neq 0$. Therefore $x \cdot y = \frac{ac}{bd}$ is rational, so a contradiction, since we assumed $x \cdot y$ is irrational. ■

Proof of Theorem 1:

Let $Q(x) :=$ "x is irrational." The Lemma establishes that Q satisfies the hypothesis of part (a): for any $x, y \in \mathbb{R}$, if $Q(x \cdot y)$ is true, then at least one of $Q(x)$ or $Q(y)$ is true. Therefore, by part (a), for any $n \geq 1$ and any $x_1, \ldots, x_n \in \mathbb{R}$, if $Q(x_1 \cdot x_2 \cdots x_n)$ is true (i.e., the product is irrational), then at least one of $Q(x_1), \ldots, Q(x_n)$ is true (i.e., at least one of the numbers is irrational). This proves Theorem 1. ■